

Factor-Contrastive Sparse Autoencoders: Controlled Relations for Stable Multilingual Features

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Abstract

Sparse autoencoders can reconstruct activations accurately while assigning a feature to the wrong source of variation: in multilingual data, a feature may follow language rather than meaning. Reconstruction and sparsity specify how compactly an activation is encoded, but not what should remain stable. We introduce the Factor-Contrastive Sparse Autoencoder, which organizes one overcomplete dictionary through reciprocal controlled relations. One sparse route preserves intent while locale changes; the other receives the reversed relation. Matched opposing examples rule out a locale-only shortcut, while blockwise BatchTopK gives both routes independent activity budgets and an ordinary decoder preserves reconstruction. On held-out MASSIVE languages, reciprocal supervision raises intent concept AUC from .701 to .912 and cross-locale stability from .150 to .225 over an architecture-matched reconstruction control. The advantage persists across dictionary widths and transfers across a second multilingual dataset and language-model family. Validation-selected features reach .817 intent purity and .725 reliable coverage. Controlled relations can therefore determine which factor organizes a sparse dictionary without assuming statistical independence or abandoning reconstruction.

1 Introduction

Sparse autoencoders (SAEs) expose language-model activations as sparse combinations of learned dictionary elements (Huben et al., 2024; Gao et al., 2025; Lieberum et al., 2024). Their features are most useful when meaning survives irrelevant changes in wording or language. Yet an SAE trained only for reconstruction may follow whichever correlated pattern best reduces error. Intent and locale, for example, both help reconstruct a multilingual activation, although an intent feature should respond across locales rather than to

a locale shared by unrelated utterances. Neither reconstruction nor sparsity expresses this preference. Standard SAE objectives therefore determine capacity and activity without determining which observed factor organizes the learned coordinates; feature quality remains distinct from reconstruction quality (Karvonen et al., 2025).

Our central idea is to specify feature organization through relations between examples. For an intent-oriented feature, a positive keeps intent fixed while changing locale; an opposing example keeps locale fixed while changing intent. Locale alone cannot solve this comparison because it favors the opposing example. Reversing the relation defines a locale-oriented route. The two relations state what each route should preserve without assuming statistical independence.

The Factor-Contrastive SAE applies these relations directly to an overcomplete sparse code. A shared encoder produces intent- and locale-dominant routes, blockwise BatchTopK (Bussmann et al., 2024) reserves a mean activity budget for each, and a linear decoder retains ordinary reconstruction. The relational loss changes feature organization without replacing the dictionary or SAE objective. An architecture-matched control removes only this loss, isolating its effect.

We test concept recovery, stability across held-out languages, reconstruction, and transfer across widths, datasets, and model families. Reciprocal supervision consistently improves feature organization over the exact reconstruction control; Section 4 reports the full comparisons.

Contributions.

- **Relation-defined factor organization.** Reciprocal matched relations specify which observed factor should dominate each sparse route and exclude a nuisance-only solution.
- **A direct sparse implementation.** Blockwise

BatchTopK realizes both routes within one overcomplete dictionary while retaining ordinary reconstruction and an exact capacity-matched control.

- **Feature-level evidence.** Paired experiments across datasets, model families, and dictionary widths connect the relational objective to concept recovery, cross-language stability, and held-out interpretability.

2 Related Work

Sparse feature learning and evaluation. SAEs approximate activations with sparse dictionaries, including large-scale dictionaries across layers and model sizes. Sparsity mechanisms govern how this capacity is used: Top- k fixes per-example activity, BatchTopK fixes mean minibatch activity, and Matryoshka SAEs optimize reconstruction at nested dictionary prefixes (Kusupati et al., 2022; Karvonen et al., 2025). Evaluation work separately measures reconstruction and feature quality. These methods control capacity, activity, or reconstruction hierarchy, but leave feature orientation to correlations in the training activations.

Structured sparse autoencoders. Relational objectives can structure sparse features beyond reconstruction. T-SAE uses token adjacency to favor temporal consistency (Bhalla et al., 2026); geometry-invariant SAEs align features across languages to recover shared reasoning structure (Bogdanov and Huang, 2026). Our target is different: reciprocal relations assign two observed factors to separate activity-controlled routes within one dictionary. The object of evaluation is therefore factor orientation rather than temporal coherence or global alignment.

Contrastive factor isolation. Contrastive learning defines representation content through positive and opposing relations (van den Oord et al., 2018; Zimmermann et al., 2021). Such structure is necessary because unsupervised factor decomposition is not identifiable without assumptions (Locatello et al., 2019); controlled transformations can specify what changes and what remains invariant (von Kügelgen et al., 2021; Cai et al., 2024). Factor-Contrastive SAE applies this principle to sparse coordinates: nuisance-matched opposition blocks a nuisance-only solution, and the reversed relation organizes the second route around that nuisance.

3 Factor-Contrastive Sparse Autoencoder

The method uses observed relations to determine how a sparse dictionary should organize its features. A relation sampler first states what each route should treat as equivalent and what it must distinguish. A blockwise sparse autoencoder then learns these relations while reconstructing the original activation. Figure 1 summarizes this two-stage design.

3.1 Controlled relations define route semantics

Let $H(x) \in \mathbb{R}^d$ be the frozen pooled activation of utterance x . Each utterance has an intent C and a locale S . The intent route z_C should respond similarly when intent is fixed and locale changes. The locale route z_S should exhibit the reciprocal behavior.

The key comparison is easiest to see from one anchor x_a with intent c_a and locale s_a . For z_C , the positive keeps the anchor intent but changes locale. The opposing example keeps the anchor locale but changes intent. Formally,

$$\begin{aligned} C(x_C^+) &= c_a, & S(x_C^+) &\neq s_a, \\ C(x_C^-) &\neq c_a, & S(x_C^-) &= s_a. \end{aligned} \quad (1)$$

This pairing makes the desired distinction explicit: the positive changes the nuisance factor, whereas the opposing example matches it. The z_S route reverses the same pair, using $x_S^+ = x_C^-$ and $x_S^- = x_C^+$. It therefore preserves locale while opposing intent.

On MASSIVE, half of the z_C positives are exact translations and half are different semantic IDs with the same intent. Every opposing example has a different intent and the anchor locale. The 50/50 mixture prevents the intent relation from reducing to sentence identity; sampling is balanced over the complete relation set.

Proposition 1. *If negatives for z_C are matched on S , a representation that depends only on S cannot rank x_C^+ above x_C^- for every relation.*

An S -only representation maps x_a and x_C^- to the same value, making the opposing similarity maximal, while x_C^+ changes S . It therefore cannot satisfy every z_C relation. Appendix A.3 gives the formal argument; the reciprocal construction gives the same protection to z_S .

175 3.2 Learning the relations in a sparse 176 dictionary

177 The relations supervise sparse features directly
178 rather than first creating a dense bottleneck. A
179 shared encoder maps $H(x)$ into one overcomplete
180 pre-activation vector and assigns disjoint coordi-
181 nates to the two routes:

$$182 \quad \begin{aligned} a(x) &= W_{\text{enc}}H(x) + b_{\text{enc}}, \\ a &= [a_C; a_S] \in \mathbb{R}^m. \end{aligned} \quad (2)$$

183 A single global sparsity budget could allow one
184 route to consume nearly all active features. We in-
185 stead apply BatchTopK separately to the two blocks
186 (Bussmann et al., 2024):

$$187 \quad \begin{aligned} z_C &= \text{BatchTopK}_{k_C}(a_C), \\ z_S &= \text{BatchTopK}_{k_S}(a_S). \end{aligned} \quad (3)$$

188 For a batch of B examples, the blocks retain their
189 largest Bk_C and Bk_S positive activations. Each
190 example may use more or fewer features, but each
191 route receives a guaranteed mean activity budget.
192 Training-calibrated thresholds make test-time infer-
193 ence independent of batch composition.

194 The exact blockwise control uses the same en-
195 coder, route widths, activity budgets, decoder, and
196 optimizer but removes both relational losses. Its
197 difference from the proposed model therefore iso-
198 lates the effect of controlled relations rather than
199 the block allocation.

200 The decoder reconstructs the original activation
201 from the concatenated code:

$$202 \quad \hat{H} = W_{\text{dec}}[z_C; z_S] + b_{\text{dec}}, \quad \mathcal{L}_{\text{rec}} = \|H - \hat{H}\|_2^2. \quad (4)$$

203 Following standard Top- k SAE training (Gao et al.,
204 2025), decoder columns are normalized after every
205 optimization step.

206 The relational objective asks each route to rank
207 its positive above its matched opposing example.
208 For route $r \in \{C, S\}$, let $\bar{z}_r = z_r / \|z_r\|_2$, $u_r^+ =$
209 $\bar{z}_r(x_a)^\top \bar{z}_r(x_r^+)$, and $u_r^- = \bar{z}_r(x_a)^\top \bar{z}_r(x_r^-)$. We
210 use the standard contrastive softmax form (van den
211 Oord et al., 2018; Zimmermann et al., 2021):

$$212 \quad \begin{aligned} \mathcal{L}_r &= -\log \frac{\exp(u_r^+ / \tau)}{\exp(u_r^+ / \tau) + \exp(u_r^- / \tau)}, \\ \mathcal{L} &= \mathcal{L}_{\text{rec}} + \frac{\lambda}{2}(\mathcal{L}_C + \mathcal{L}_S). \end{aligned} \quad (5)$$

213 The first term preserves ordinary SAE reconstruc-
214 tion; the two relational terms determine which fac-
215 tor organizes each route. No adversarial, orthogo-
216 nality, independence, prototype, or swap-training
217 term is used.

The main model uses a fourfold dictionary; an 218
eightfold version tests sparse-width sensitivity. Ap- 219
pendix A gives the exact route dimensions and 220
activity budgets. Complementarity is evaluated 221
from the learned routes rather than imposed by an 222
auxiliary constraint. 223

4 Experimental Evaluation 224

The experiments ask four questions. Do controlled 225
relations improve intent-oriented features over an 226
identical reconstruction-only model? Do these 227
gains remain when reconstruction and activity are 228
measured? Are the learned features stable and in- 229
terpretable across held-out locales? Finally, do the 230
conclusions transfer across sparse widths, datasets, 231
and model families? Appendix A contains the com- 232
plete protocol and configuration. 233

4.1 Experimental design 234

Models and data. The main experiment uses 235
masked-mean hidden-state-8 activations from 236
Gemma 2 2B (Gemma Team, 2024) on MASSIVE 237
(FitzGerald et al., 2023). Training contains 10,343 238
semantic IDs from 49 locales; validation contains 239
1,149 disjoint IDs. The final test set contains 2,968 240
untouched IDs, each observed in Arabic and Chi- 241
nese. Transfer experiments use MTOP (Li et al., 242
2021) and Pythia-160M (Biderman et al., 2023). 243

Fair comparisons. We compare raw H , global 244
BatchTopK, the exact blockwise reconstruction 245
control, Matryoshka SAE reconstruction, one-sided 246
supervision, and a validation-selected triplet loss 247
(Schroff et al., 2015). Learned comparisons share 248
their inputs, dictionary capacity, activity, initializa- 249
tion, batches, optimizer, epochs, and paired seeds. 250
Thus the blockwise control differs from our model 251
only in the relational objective. 252

What a successful route should do. A useful z_C 253
feature should identify an intent, activate consis- 254
tently for its translations, and reveal less locale in- 255
formation. We measure these properties with intent 256
concept AUC, Arabic–Chinese activation correla- 257
tion, and a locale probe. Intent- and locale-oriented 258
feature fractions show which factor dominates the 259
active dictionary. The reciprocal diagnostics ask 260
whether z_S retains locale while suppressing intent. 261

Intent concept AUC is the mean one-vs-rest held- 262
out AUC (Fawcett, 2006) of one validation-selected 263
coordinate per intent. Stability is the mean Pearson 264
correlation (Pearson, 1895) between paired Arabic 265

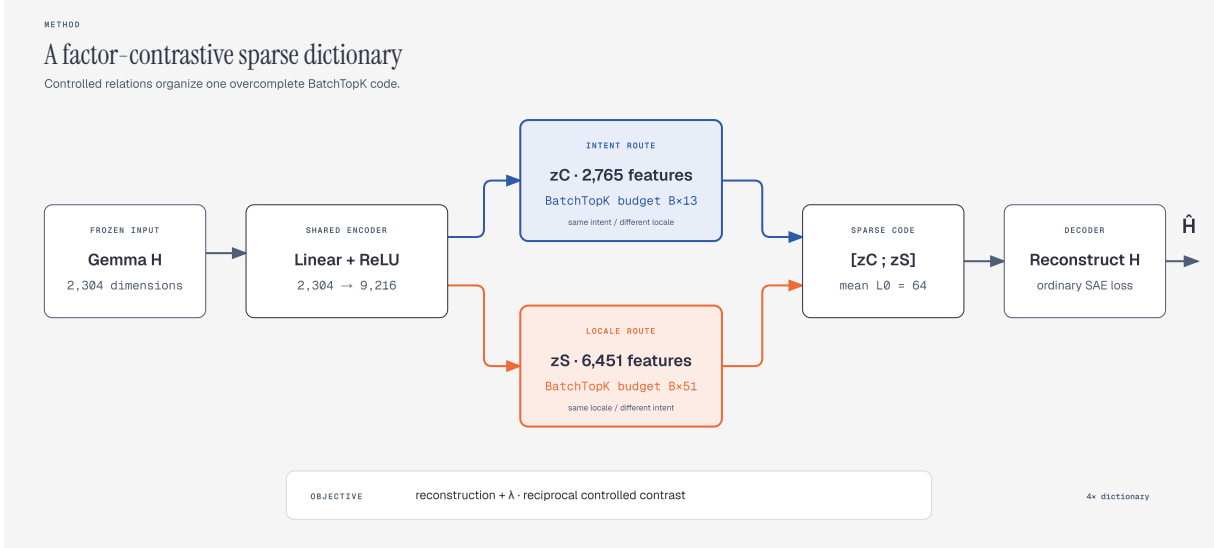


Figure 1: Factor-contrastive sparse autoencoder. A shared encoder maps frozen activations into one overcomplete dictionary. Blockwise BatchTopK assigns separate activity budgets to the intent- and locale-dominant routes; reciprocal controlled relations organize the routes while an ordinary linear decoder reconstructs the input activation.

Table 1: MASSIVE results on held-out Arabic and Chinese. Blockwise methods are evaluated on z_C in the primary columns. Lower locale probe, locale-oriented fraction, and z_S intent probe are better.

Method	Intent AUC $\uparrow$	Stability $\uparrow$	Locale probe $\downarrow$	Intent frac. $\uparrow$	Locale frac. $\downarrow$	z_S locale $\uparrow$	z_S intent $\downarrow$	FVE $\uparrow$
Raw H	.7703	.3013	.9997	.1159	.7409	–	–	–
BatchTopK SAE	.7940 $\pm$.0014	.1278 $\pm$.0011	.9871 $\pm$.0048	.4579 $\pm$.0026	.4723 $\pm$.0030	–	–	.6618 $\pm$.0017
Blockwise control	.7007 $\pm$.0109	.1501 $\pm$.0080	.9665 $\pm$.0106	.4830 $\pm$.0122	.4465 $\pm$.0122	.9878 $\pm$.0014	.4798 $\pm$.0151	.6627 $\pm$.0002
Matryoshka SAE	.8352 $\pm$.0150	.1385 $\pm$.0017	.9821 $\pm$.0009	.4686 $\pm$.0018	.4652 $\pm$.0033	–	–	.6482 $\pm$.0014
One-sided factor SAE	.9165 $\pm$.0083	.1813 $\pm$.0061	.8969 $\pm$.0143	.6923 $\pm$.0198	.2427 $\pm$.0163	.9861 $\pm$.0029	.4679 $\pm$.0004	.6415 $\pm$.0015
Reciprocal factor SAE (ours)	.9124 $\pm$.0027	.2248 $\pm$.0035	.8944 $\pm$.0034	.7864 $\pm$.0144	.1646 $\pm$.0065	.9925 $\pm$.0008	.3892 $\pm$.0079	.6098 $\pm$.0011
Triplet, $m = .2$	.9063 $\pm$.0190	.1983 $\pm$.0029	.9096 $\pm$.0049	.7274 $\pm$.0013	.2043 $\pm$.0011	.9916 $\pm$.0029	.3818 $\pm$.0171	.6427 $\pm$.0012

and Chinese activations. Balanced logistic probes (Cox, 1958) measure locale in z_C and locale or intent in z_S . We report reconstruction as fraction of variance explained (FVE) and sparsity as the observed number of active features.

Every method uses the same feature-selection split, probe split, test split, and evaluator. Features are selected only on validation data, and all comparison tables report mean $\pm$ standard deviation over three paired seeds. Appendix A defines feature activity, orientation, probe selection, and bootstrap estimation.

Validation before testing. Validation selects the reciprocal binary objective and triplet margin $m = .2$ from $\{.1, .2, .4\}$. The Arabic–Chinese test activations are then evaluated once. No test result selects a model, feature, intent, or displayed example.

4.2 Controlled relations reorganize the routes

Controlled relations reorganize both routes in the intended direction. Against the exact blockwise control, our z_C raises intent AUC from .7007 to

.9124 and stability from .1501 to .2248. Its intent-oriented feature fraction rises from .4830 to .7864, while its locale-oriented fraction falls from .4465 to .1646. The reciprocal z_S route reaches .9925 locale accuracy and reduces intent leakage from .4798 to .3892. Because the control removes only the relational losses, these changes isolate relational supervision.

The same conclusion holds against the standard sparse alternatives. Relative to global BatchTopK, intent AUC increases by .1185 and stability by .0970; relative to Matryoshka, the gains are .0772 and .0863. The complete-route intent probe remains close to global BatchTopK (.4625 versus .4680), indicating that stronger organization in z_C does not remove intent from the joint sparse code.

Raw H is a dense reference, not a sparse base line. Locale is almost perfectly decoded from it (.9997), and 74.1% of its coordinates are locale-oriented. In contrast, the learned z_C is sparse and overcomplete, with 78.6% intent-oriented and 16.5% locale-oriented active features. The result is therefore a change in feature organization, not

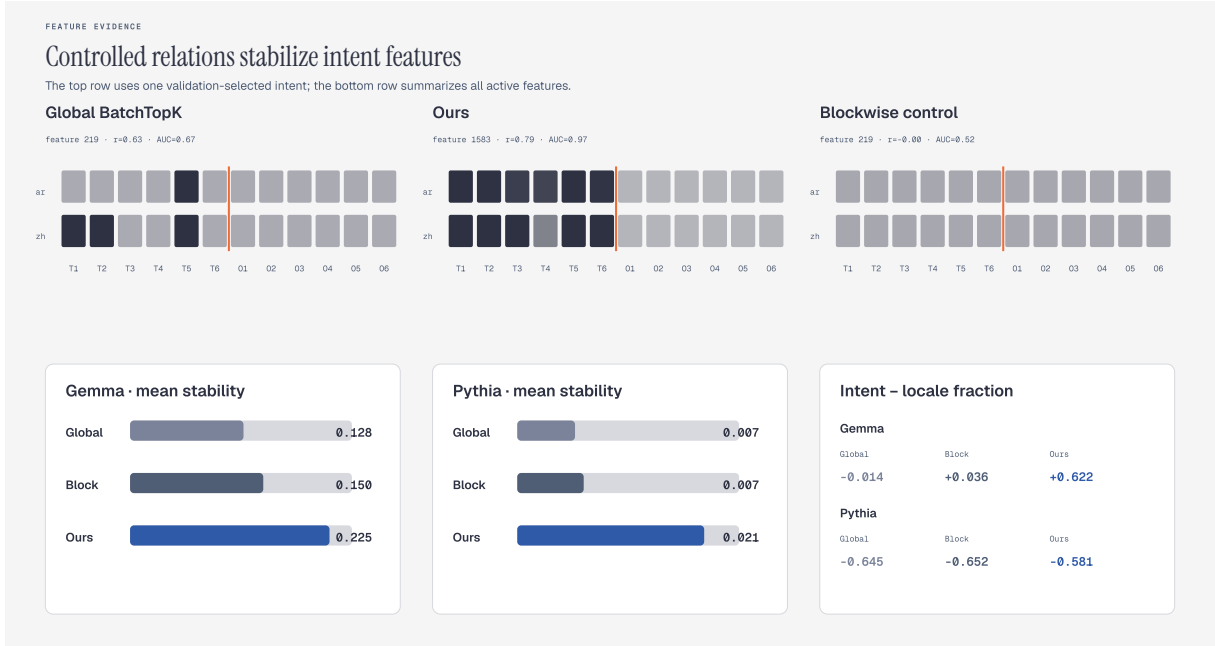


Figure 2: Controlled relations stabilize intent features. Top: each Gemma method selects its own feature for the same validation-selected intent; T1–T6 and O1–O6 are held-out target- and comparison-intent semantic IDs in Arabic and Chinese. Bottom: three-seed all-feature stability and intent-minus-locale orientation for Gemma and Pythia.

simply the presence of intent somewhere in the representation.

4.3 Reconstruction and activity

The factor gains are obtained at matched sparse activity with a moderate reconstruction trade-off. All models train at mean $L_0 = 64$, and their frozen thresholds produce similar held-out activity: $104.46 \pm .07$ for ours, $105.99 \pm .81$ for the exact control, and $107.16 \pm .91$ for global BatchTopK. Reconstruction FVE is .6098 for ours and .6627 for the exact control, an 8.0% relative difference.

Held-out L_0 exceeds the training budget because test examples use thresholds calibrated on seen locales rather than a newly computed test-batch budget. Since ours and the controls have nearly identical held-out activity, the AUC and stability gains cannot be explained by a larger test-time code. Appendix A gives the shared thresholding and reconstruction protocol.

4.4 Cross-locale feature consistency

Figure 2 shows what the aggregate stability gain looks like for one intent feature. The reciprocal feature responds similarly to Arabic and Chinese instances of its target intent, reaching held-out AUC .97 and stability .79. The blockwise-control feature for the same validation-selected intent reaches

AUC .52 and approximately zero stability. The dictionary-wide averages give the same ordering for both Gemma (.2248 versus .1501) and Pythia (.0211 versus .0075).

Feature and intent selection use disjoint validation splits; held-out activations are never used to choose the example. Test labels only identify the six deterministically ordered target and comparison instances after selection. The complete selection rule appears in Appendix A.

4.5 Intent-feature interpretability

The stable features are also more interpretable. Reciprocal supervision raises Purity@20 by .236 and reliable coverage by .384 over the exact blockwise control. Its near-zero top-ID overlap indicates that different intent features retrieve distinct examples rather than the same globally active utterances.

Purity ranks unique semantic IDs by the larger activation across their Arabic and Chinese translations. The calculation covers the 46 intents with at least 20 held-out IDs; reliable coverage is the fraction whose selected feature reaches at least .8 purity. Selected-feature stability covers all 58 intents.

The normalized locale entropy of our top activation sets is .633, close to global BatchTopK (.636), so high purity does not come from retrieving only

Table 2: Validation-selected intent features on held-out MASSIVE Arabic and Chinese. Purity and reliable coverage use 46 intents with at least 20 test semantic IDs; stability covers all 58 intents.

Method	Purity@20 $\uparrow$	Reliable coverage $\uparrow$	Selected stability $\uparrow$	Top-ID overlap $\downarrow$
Global BatchTopK	.653 $\pm$.020	.370 $\pm$.022	.457 $\pm$.004	.0025 $\pm$.0005
Blockwise control	.581 $\pm$.068	.341 $\pm$.091	.375 $\pm$.040	.0034 $\pm$.0012
Reciprocal factor SAE (ours)	.817 $\pm$.012	.725 $\pm$.045	.563 $\pm$.003	.0022 $\pm$.0001



Figure 3: Validation-selected sparse features recover recognizable intent concepts on held-out Arabic and Chinese. The cards show representative high-activation examples for three intent features; selection uses validation data only, while purity and stability are measured on the held-out locales.

one held-out locale. Appendix D provides the complete 58-intent catalogue and qualitative examples. The complete representative-seed catalogue contains all 58 intent features, including feature IDs, purity, stability, held-out AUC, and activity support (Appendix D). The machine-readable catalogue extends this evaluation to all three methods and seeds.

4.6 Robustness, transfer, and ablations

The controlled-relation advantage persists when the dictionary is widened, exact translation IDs are removed, or the activation model is changed. Figure 4 summarizes these three tests; the tables below and Appendix C give the full results.

Doubling dictionary width and activity preserves the main ordering. At eightfold width, reciprocal supervision improves AUC by .1783 and stability by 25.6% over the exact control. FVE remains within 6.3% of the control, and intent AUC is nearly unchanged across widths (.9124 and .9156).

MTOP tests whether exact translations are necessary. Training uses German, English, Spanish,

and French, while Hindi and Thai are held out. Because MTOP has no parallel semantic IDs, stability compares each feature’s 50-intent activation profile across the two test languages. Reciprocal supervision improves AUC by .2748, stability by .1277, relation margin by .3214, and retrieval R@1 by .2910 over the exact blockwise control. The same objective therefore transfers without exact translations or dataset-specific architectural changes.

Pythia-160M tests a different activation geometry. Reciprocal supervision raises stability from .0075 to .0211 and improves relation margin by .4797 over the blockwise control, with the same direction in all three paired seeds. It also improves feature orientation, locale leakage, and the opposing route. The absolute AUC change is small in this weaker activation space; stability, margin, and route organization provide the transfer evidence. No Pythia-specific loss or sparsity setting is introduced.

The component ablations support reciprocal binary supervision. One-sided training attains

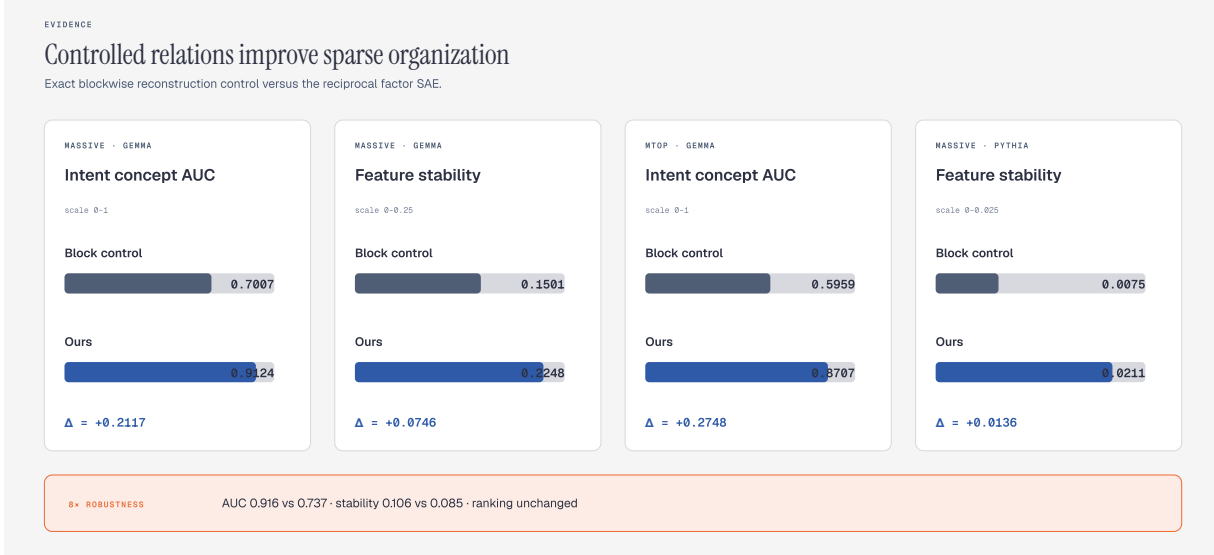


Figure 4: The controlled-relation advantage transfers across datasets and model families and persists at eightfold width. All panels compare against the exact blockwise reconstruction control; the Pythia table and complete protocol are in Appendix C.

Table 3: MASSIVE width robustness. Dictionary width and activity double together from $4 \times / L_0 = 64$ to $8 \times / L_0 = 128$.

Width	Method	AUC ↑	Stability ↑	Locale probe ↓	Intent frac. ↑	Locale frac. ↓	FVE ↑
4×	Global BatchTopK	.7940	.1278	.9871	.4579	.4723	.6618
4×	Blockwise control	.7007	.1501	.9665	.4830	.4465	.6627
4×	Ours	.9124	.2248	.8944	.7864	.1646	.6098
8×	Global BatchTopK	.8094	.0797	.9908	.4408	.4637	.7152
8×	Blockwise control	.7373	.0847	.9760	.4494	.4562	.7154
8×	Ours	.9156	.1063	.9397	.6708	.2530	.6705

slightly higher intent AUC (.9165 versus .9124), but reciprocal training improves stability by .0434, raises the intent-oriented fraction by .0941, lowers the locale-oriented fraction by .0781, and reduces z_S intent leakage by .0787. The triplet loss retains more reconstruction FVE (.6427 versus .6098) but is lower on AUC, stability, locale leakage, and feature orientation. These results motivate the reciprocal binary objective used throughout the transfer experiments.

5 Discussion and Conclusions

Controlled relations change feature organization. The exact blockwise control is the key comparison because it shares every architectural and optimization choice with the proposed model. Its gap to the reciprocal model therefore measures the effect of the controlled relations. On MASSIVE, reciprocal supervision increases intent AUC, cross-locale stability, and the intent-oriented fea-

ture fraction while decreasing the locale-oriented fraction. The reciprocal z_S result shows that this change is not obtained by simply suppressing the second route: the opposing block retains locale and loses intent information in the intended direction.

The result is not tied to one sparse scale. The same ordering holds when dictionary width and activity are doubled. Absolute coordinate-level stability decreases at eightfold width because evidence is distributed across more active coordinates, but the relational advantage remains and intent AUC is unchanged. The fourfold model is therefore the compact operating point rather than a uniquely favorable width.

Relational supervision transfers. MTOP re-moves the exact translation identifiers available in MASSIVE, yet same-intent/different-language relations improve AUC, margin, retrieval, and stability on held-out Hindi and Thai. Pythia-160M changes activation dimensionality and model fam-

Table 4: MTOP transfer to held-out Hindi and Thai (three paired seeds). Lower language probe and z_S intent are better.

Method	AUC $\uparrow$	Stability $\uparrow$	Margin $\uparrow$	Lang. probe $\downarrow$	R@1 $\uparrow$	z_S lang. $\uparrow$	z_S intent $\downarrow$
Raw H	.7859	.5443	-.0724	1.0000	.7998	–	–
Global BatchTopK	.6529 $\pm$.0038	.3605 $\pm$.0023	.0341 $\pm$.0024	.9837 $\pm$.0020	.7010 $\pm$.0062	–	–
Blockwise control	.5959 $\pm$.0186	.3701 $\pm$.0119	.0228 $\pm$.0100	.9416 $\pm$.0050	.4900 $\pm$.0391	.9795 $\pm$.0027	.5071 $\pm$.0140
Ours	.8707 $\pm$.0017	.4978 $\pm$.0153	.3441 $\pm$.0105	.9278 $\pm$.0034	.7809 $\pm$.0166	.9857 $\pm$.0035	.4639 $\pm$.0062

ily while preserving the sparse capacity and ob-
jective. The consistent direction indicates that the
effect follows the controlled relations rather than
one model-specific activation geometry.

Scope and conclusion. The method requires ob-
served factors from which controlled relations can
be constructed; the present evidence concerns mul-
tilingual intent and language. It targets factor-
dominant routes rather than coordinate-wise statis-
tical independence. Within this setting, the results
support a simple design principle: when a desired
invariance can be expressed as a matched relation,
it can be imposed directly on sparse feature learn-
ing while retaining the standard reconstruction ob-
jective.

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602 A Frozen Experimental Protocol

603 A.1 MASSIVE split and relation audit

604 The MASSIVE split is fixed by semantic ID. Representation training contains 10,343 IDs, validation
605 contains 1,149 disjoint IDs, and the untouched Arabic–Chinese test set contains 2,968 IDs. Feature
606 selection uses 5,220 balanced validation rows spanning all 58 intents and 49 seen locales; its fit and
607 scoring semantic IDs are disjoint. Probe fitting uses 5,800 balanced training rows. The canonical manifest
608 hash is 6c33455d91ccd15fc6054c3f077cb2393f387ab136db71fcea76f581d7344439.

609 For z_C , half of sampled positives are exact translations and half use a different semantic ID with the
610 same intent. Every opposing example has a different intent and the anchor locale. The aggregate schedule
611 is exactly balanced; intents with only one semantic ID necessarily contribute exact translations, and
612 eligible anchors from other intents compensate with different-ID positives. The z_S route reverses the
613 assignments.

614 A.2 Deterministic sampler and evaluator

615 The sampler and evaluator follow one frozen sequence for every learned comparison:

- 616 1. sample an anchor from the fixed training-ID manifest;
- 617 2. with probability .5, choose its exact translation as the z_C positive; otherwise choose a different
618 semantic ID with the same intent in a different locale;
- 619 3. choose the z_C opposing example with a different intent in the anchor locale, and reverse these
620 assignments for z_S ;
- 621 4. apply global or blockwise BatchTopK according to the comparison while keeping the same batch
622 order and mean activity budget;
- 623 5. select one coordinate per intent using only the validation fit split, score selection on the disjoint
624 validation score split, and freeze the selected coordinates;
- 625 6. evaluate the frozen model and selected coordinates once on the held-out locale test split.

626 Batch ordering and relation sampling use separate seeded random generators. Ties in ranking metrics
627 follow stable row order, and every method uses the same active-feature, variance-floor, probe, and
628 bootstrap implementations.

629 A.3 Shortcut proposition

630 Let similarity be cosine similarity and suppose $z_C(x) = f(S(x))$. Since $S(x_a) = S(x_C^-)$, $z_C(x_a) =$
631 $z_C(x_C^-)$ and therefore $u_C^- = 1$ whenever the code is nonzero. Cosine similarity is at most one, so $u_C^+ > u_C^-$
632 is impossible. Consequently, an S -only representation cannot satisfy the strict ranking required by every
633 matched relation. The argument does not require statistical independence between C and S ; it uses only
634 the matched construction in Equation 1.

635 A.4 Metric definitions

636 Intent concept AUC is the mean one-vs-rest held-out AUC of one validation-selected coordinate per
637 intent. Cross-locale stability is the mean Pearson correlation of paired Arabic–Chinese feature activations
638 across active coordinates. An intent-oriented feature has intent selectivity greater than 1.1 times locale
639 selectivity; locale orientation is defined symmetrically. Locale leakage is balanced locale-probe accuracy
640 on z_C . Opposing-route diagnostics report locale accuracy and intent balanced accuracy (Brodersen et al.,
641 2010) on z_S . Probe coordinates are selected by training-only one-way ANOVA (Fisher, 1925) before the
642 standardized logistic model is fitted.

643 Interpretability selects one feature per intent on the disjoint validation fit split. Held-out Arabic and
644 Chinese translations are merged by semantic ID using the larger activation, then ranked. Purity@20 is
645 the fraction of the first 20 unique IDs carrying the selected intent. Reliable coverage is the fraction of

eligible intents whose purity is at least .8. The headline calculation contains the 46 intents with at least 20 held-out IDs.

All feature metrics use variance floor 10^{-12} , activation threshold 10^{-8} , and minimum activity rate 10^{-3} . Bootstrap confidence intervals use 10,000 resamples (Efron and Tibshirani, 1993). Tables report mean $\pm$ standard deviation over seeds 20260827, 20260828, and 20260829.

A.5 Training configuration

All learned comparisons use AdamW (Loshchilov and Hutter, 2019). Table 5 gives the exact dimensions, route budgets, optimization settings, and training duration for the main experiment.

Table 5: Frozen fourfold MASSIVE configuration.

Choice	Value
Activation source	Gemma 2 2B, hidden state 8
Input / dictionary width	2,304 / 9,216
z_C/z_S widths	2,765 / 6,451
BatchTopK budgets	13B/51B
Mean training L_0	64
Batch size / epochs	128 / 30
Optimizer / learning rate	AdamW / 10^{-4}
Temperature τ / weight λ	.07 / 1

At eightfold width, the dictionary contains 18,432 coordinates and uses budgets 26B/102B, preserving route proportions and active fraction. MTOP uses the main fourfold configuration. Pythia-160M uses 768-dimensional hidden-state-8 activations while retaining the 9,216-coordinate dictionary and 13B/51B budgets. Training was performed on a single NVIDIA RTX 3090 with 24GB memory.

A.6 Layer sensitivity

The main experiments use Gemma hidden state 8. To test whether the result depends on that choice, we repeat the decisive paired comparison at hidden states 4, 8, 16, and 24. Each layer receives its own training-set normalization, and the proposed model and exact blockwise control start from the same fresh initialization. The manifest, sampler, dictionary, activity budgets, optimizer, training duration, and evaluator are otherwise unchanged. This appendix experiment uses seed 20260827; the main hidden-state-8 result in Table 1 remains the three-seed estimate.

Table 6: Gemma layer sensitivity. Each entry is exact blockwise control / ours from a paired run. The proposed relations improve intent AUC and cross-locale stability and reduce locale-probe accuracy at all four layers.

Hidden state	Intent AUC $\uparrow$	Stability $\uparrow$	Locale probe $\downarrow$	FVE $\uparrow$	Held-out L_0
4	.6225 / .8989	.0277 / .0594	.9720 / .9053	.4879 / .4367	109.13 / 105.03
8	.7132 / .9097	.1509 / .2184	.9639 / .8908	.6631 / .6098	107.57 / 105.45
16	.6847 / .8956	.1613 / .2223	.9879 / .9464	.7222 / .6680	117.40 / 118.60
24	.6771 / .9173	.0857 / .1463	.9808 / .9609	.6392 / .6030	139.18 / 132.72

The strongest absolute AUC occurs at hidden state 24, whereas hidden state 8 gives the lowest locale probe and nearly the highest stability. More importantly, the paired direction is invariant to layer choice: AUC improves by .1965–.2764, stability by .0317–.0675, and locale-probe accuracy falls by .0199–.0731. The reconstruction cost is modest and consistent, with FVE decreasing by .0362–.0542.

A.7 Software and computation

The fourfold Gemma/MTOP SAE has 42,478,848 trainable encoder/decoder parameters, the eightfold model has 84,955,392, and the Pythia model has 14,165,760. The final study comprises 53 thirty-epoch

Table 7: Frozen software and execution environment. Versions are taken from the submitted `uv.lock`.

Component	Version or setting
Python	$\geq 3.12, < 3.14$
PyTorch / CUDA build	2.7.1 / cu128
Transformers / Datasets	5.15.1 / 5.0.1
NumPy / pandas	2.5.2 / 3.0.5
scikit-learn / Accelerate	1.9.0 / 1.14.0
SAE Lens	6.49.1
Hardware	NVIDIA RTX 3090, 24GB
Gemma extraction	bfloat16 model, float32 pooled arrays
Pythia extraction	float16 model, float32 pooled arrays
Tokenization / pooling	128 tokens / masked non-padding mean

training runs: 18 MASSIVE main comparisons, nine eightfold runs, nine MTOP runs, nine Pythia runs, and eight paired layer-sensitivity runs, in addition to the frozen validation candidates. Saved checkpoints are approximately 170MB, 340MB, and 56.7MB for the fourfold Gemma/MTOP, eightfold Gemma, and Pythia configurations, respectively.

B Comparison and Selection Details

Global BatchTopK uses one dictionary and a global $64B$ activity budget. The exact blockwise control has the proposed encoder, decoder, 30/70 route widths, and $13B/51B$ budgets but optimizes reconstruction only. The Matryoshka SAE baseline (Kusupati et al., 2022; Karvonen et al., 2025) averages the bias-only reconstruction with prefix reconstructions at 576, 1,152, 2,304, 4,608, and 9,216 features. The one-sided ablation applies the controlled loss only to z_C . The triplet ablation (Schroff et al., 2015) replaces the binary loss with $\max(0, m + u^- - u^+)$; validation selects $m = .2$ from $\{.1, .2, .4\}$.

On the frozen validation split, the reciprocal binary objective has the highest sparse-feature stability (.3921) and lowest locale probe (.8373) among candidate SAEs. All learned comparisons use paired initialization, identical normalized inputs, batches, optimizer, epochs, and mean training activity. Raw H is a dense-coordinate reference rather than an SAE baseline.

Table 8: Frozen MASSIVE validation selection. Lower locale probe and locale-oriented fraction are better; all other feature metrics increase. The reciprocal binary objective is selected by stability and locale suppression; $m = .2$ is the retained triplet control.

Method	AUC	Stability	Locale probe	Intent frac.	Locale frac.	FVE
Global BatchTopK	.7264	.2300	.9712	.5068	.3914	.7867
Blockwise control	.6663	.2683	.9051	.5434	.3546	.7866
Matryoshka SAE	.7568	.2496	.9737	.5197	.3720	.7742
One-sided	.8845	.3239	.8390	.7162	.1903	.7680
Reciprocal binary	.8723	.3921	.8373	.7814	.1387	.7467
Triplet, $m = .1$	.8288	.3622	.8958	.7137	.1804	.7753
Triplet, $m = .2$	.8616	.3537	.8653	.7348	.1703	.7739
Triplet, $m = .4$	.8625	.3183	.8746	.7595	.1515	.7696

C Dataset and Model Transfer

For MTOP, German, English, Spanish, and French are used for training; Hindi and Thai are held out. Training contains 10,213 rows, feature selection contains 800 disjoint rows, and test contains 5,261

Table 9: Paired per-seed results for the decisive comparison. Each entry is exact blockwise control / reciprocal model. A dash denotes a metric not used in that setting.

Setting	Seed	AUC	Stability	Locale probe	Margin	FVE	Held-out L_0
MASSIVE 4×	20260827	.7132/.9127	.1569/.2208	.9582/.8979	–	.6627/.6107	106.52/104.39
	20260828	.6938/.9150	.1522/.2275	.9784/.8912	–	.6625/.6086	106.40/104.47
	20260829	.6951/.9097	.1413/.2260	.9629/.8942	–	.6628/.6102	105.06/104.53
MASSIVE 8×	20260827	.7191/.9213	.0837/.1052	.9747/.9276	–	.7149/.6707	226.12/226.21
	20260828	.7559/.9127	.0856/.1067	.9757/.9447	–	.7156/.6707	224.51/222.96
	20260829	.7369/.9128	.0848/.1071	.9774/.9468	–	.7157/.6702	224.84/224.41
MTOP	20260827	.6131/.8696	.3677/.4951	.9445/.9266	.0288/.3331	.1412/.1201	64/64
	20260828	.5985/.8726	.3830/.4840	.9445/.9251	.0283/.3455	.1408/.1203	64/64
	20260829	.5761/.8698	.3596/.5143	.9358/.9316	.0112/.3538	.1422/.1230	64/64
Pythia-160M	20260827	.5216/.5264	.0096/.0198	.9848/.9788	–.5888/ –.0819	.7583/.7062	64/64
	20260828	.5183/.5163	.0068/.0228	.9865/.9791	–.5268/ –.0756	.7574/.7096	64/64
	20260829	.5138/.5253	.0060/.0206	.9933/.9798	–.5874/ –.1064	.7588/.7022	64/64

rows over 50 intents. Because MTOP lacks parallel semantic IDs, stability is the Pearson correlation between each feature’s 50-intent mean-activation profiles in Hindi and Thai. The MTOP manifest hash is 7e3f0c13e42fa7016d0d2d678335eed7ae5cdb73549daa13d28d4cba7ded83e3.

For model-family transfer, Pythia-160M replaces Gemma while the MASSIVE manifest, sampler, sparse capacity, budgets, optimizer, epochs, and seeds remain fixed. Activations are masked-mean hidden state 8; the complete result appears in Table 10.

Table 10: MASSIVE model-family transfer with Pythia-160M (three paired seeds). Lower locale probe, locale fraction, and z_S intent are better.

Method	AUC ↑	Stability ↑	Margin ↑	Locale probe ↓	Intent frac. ↑	Locale frac. ↓	z_S locale ↑	z_S intent ↓
Raw H	.5824	.0737	-.4487	1.0000	.0521	.9180	–	–
Global BatchTopK	.5349 ± .0016	.0065 ± .0002	–.5614 ± .0177	.9914 ± .0008	.1493 ± .0052	.7941 ± .0062	–	–
Blockwise control	.5179 ± .0039	.0075 ± .0019	–.5677 ± .0354	.9882 ± .0045	.1497 ± .0090	.8016 ± .0085	.9907 ± .0031	.0823 ± .0056
Ours	.5227 ± .0055	.0211 ± .0016	–.0879 ± .0163	.9792 ± .0005	.1792 ± .0083	.7604 ± .0048	.9937 ± .0015	.0581 ± .0102

D Feature Examples and Complete Catalogue

The machine-readable catalogue contains all methods and seeds. Table 11 reports every intent for the representative seed 20260827.

Table 11: Complete seed-20260827 intent-feature catalogue for the reciprocal factor SAE. Features are selected on validation data and evaluated on held-out Arabic/Chinese semantic IDs.

Intent	Feature	Purity@20	Stability	AUC	Active IDs
alarm_query	1904	0.600	-0.006	0.733	37
alarm_remove	2454	0.700	0.289	0.972	209
alarm_set	1111	0.800	0.113	0.839	55
audio_volume_down	720	0.375	-0.002	0.771	16
audio_volume_mute	992	0.850	0.652	0.982	203
audio_volume_up	2278	0.350	0.761	0.913	161
calendar_query	26	0.800	0.752	0.946	909
calendar_remove	1722	1.000	0.641	0.988	281
calendar_set	749	1.000	0.790	0.988	764
cooking_recipe	1248	1.000	0.724	0.990	248

Continued on next page

Table 11: Complete seed-20260827 intent-feature catalogue for the reciprocal factor SAE. Features are selected on validation data and evaluated on held-out Arabic/Chinese semantic IDs.

Intent	Feature	Purity@20	Stability	AUC	Active IDs
datetime_convert	2752	0.550	0.660	0.898	38
datetime_query	138	1.000	0.771	0.998	492
email_addcontact	1591	0.300	0.206	0.829	45
email_query	1412	1.000	0.750	0.952	270
email_querycontact	2122	0.800	0.669	0.956	196
email_sendemail	722	1.000	0.850	0.980	373
general_greet	2700	0.000	0.000	0.500	1
general_joke	1453	0.778	0.528	0.868	18
general_quirky	979	0.800	0.613	0.827	651
iot_cleaning	1613	0.900	0.557	0.979	100
iot_coffee	1423	1.000	0.906	0.986	47
iot_hue_lightchange	469	0.900	0.750	0.970	97
iot_hue_lightdim	2274	0.400	0.466	0.943	111
iot_hue_lightoff	1954	0.950	0.548	0.985	203
iot_hue_lighton	2135	0.000	-0.001	0.499	8
iot_hue_lightup	1227	0.750	0.644	0.923	73
iot_wemo_off	857	0.700	0.464	0.943	39
iot_wemo_on	2725	0.000	-0.000	0.500	2
lists_createoradd	1583	0.300	0.794	0.979	531
lists_query	2614	0.850	0.572	0.980	404
lists_remove	2226	0.450	0.134	0.593	33
music_dislikeness	1058	0.000	-0.001	0.497	18
music_likeness	1872	0.700	0.618	0.910	139
music_query	1233	0.850	0.519	0.948	232
music_settings	1294	0.000	0.000	0.499	6
news_query	487	1.000	0.796	0.988	690
play_audiobook	2093	1.000	0.642	0.981	634
play_game	1651	1.000	0.675	0.942	56
play_music	594	0.950	0.728	0.983	714
play_podcasts	238	1.000	0.674	0.910	134
play_radio	631	0.950	0.730	0.967	317
qa_currency	650	0.950	0.891	1.000	110
qa_definition	908	1.000	0.673	0.981	557
qa_factoid	2510	0.900	0.770	0.975	942
qa_maths	159	1.000	0.745	0.940	36
qa_stock	490	0.800	0.863	0.980	94
recommendation_events	1818	0.900	0.583	0.956	362
recommendation_locations	2523	0.550	0.704	0.993	351
recommendation_movies	1250	0.700	0.656	0.921	88
social_post	760	0.950	0.827	0.986	675
social_query	2056	0.400	0.519	0.696	40
takeaway_order	299	0.600	0.692	0.970	242
takeaway_query	1497	0.650	0.610	0.995	337
transport_query	962	0.800	0.656	0.968	495
transport_taxi	2129	0.900	0.813	0.977	83
transport_ticket	2107	0.800	0.507	0.984	80

Continued on next page

Table 11: Complete seed-20260827 intent-feature catalogue for the reciprocal factor SAE. Features are selected on validation data and evaluated on held-out Arabic/Chinese semantic IDs.

Intent	Feature	Purity@20	Stability	AUC	Active IDs
transport_traffic	2	0.650	0.325	0.963	119
weather_query	2680	0.950	0.889	0.981	384

E Reproduction Entry Points

```

uv run python code/run_massive_factor_sae.py --final 700
uv run python code/evaluate_massive_factor_sae.py --final 701
uv run python code/run_massive_factor_sae.py --final \ 702
    --width-multiplier 8 --experiment batchtopk \ 703
    --experiment block_control --experiment reciprocal 704
uv run python code/evaluate_massive_factor_sae.py --width8 705
uv run python code/run_mtop_factor_sae.py --train 706
uv run python code/evaluate_mtop_factor_sae.py --evaluate 707
uv run python code/run_pythia_massive_factor_sae.py --extract 708
uv run python code/run_pythia_massive_factor_sae.py --train 709
uv run python code/run_pythia_massive_factor_sae.py --evaluate 710
uv run python code/evaluate_feature_interpretability.py 711
uv run python code/run_massive_factor_sae_layer_sensitivity.py 712
uv run python code/build_factor_stability_figure.py 713
uv run pytest -q 714

```

Machine-readable summaries are:

- Report/factor_sae_step4_definitive_test_summary.csv 716
- Report/factor_sae_step4_definitive_test_per_seed.csv 717
- Report/factor_sae_step5_width8_test_summary.csv 718
- Report/factor_sae_step5_width8_test_per_seed.csv 719
- Report/factor_sae_step6_mtop_test_summary.csv 720
- Report/factor_sae_step6_mtop_test_per_seed.csv 721
- Report/factor_sae_pythia160m_transfer_summary.csv 722
- Report/factor_sae_pythia160m_transfer_per_seed.csv 723
- Report/factor_sae_feature_interpretability_per_seed.csv 724
- Report/factor_sae_layer_sensitivity.csv 725

The accompanying supplementary archive factor-sae.zip contains the frozen code, dependency 726 lock, deterministic manifests, protocol tests, per-seed CSVs, and complete feature catalogue. A matching 727 SHA-256 file is supplied beside the archive. 728